



COMPETENCY STANDARD

FOR

SMART SYSTEMS ENGINEERING: EMBEDDED DESIGN AND SEMICONDUCTOR APPLICATION INTERFACES

(SEMICONDUCTOR / VLSI SECTOR)

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Skills for Industry Competitiveness and Innovation Program (SICIP)

Finance Division, Ministry of Finance

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The Competency Standard for **Smart Systems Engineering: Embedded Design and Semiconductor Application Interfaces** serves as a foundational document for the development of curricula, teaching and learning materials, practical lab activities and assessment tools specifically tailored for the Semiconductor/Embedded Systems sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP) and implemented by the Bangladesh Semiconductor Industry Association (BSIA), this document addresses the skills requirements critical for Smart Systems Engineering: Embedded Design and Semiconductor Application Interfaces. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Smart Systems Engineering: Embedded Design and Semiconductor Application Interfaces course aims to develop a skilled workforce capable of designing, developing, testing and integrating embedded electronic systems with semiconductor application interfaces. The course emphasizes embedded firmware development, MCU/SoC architecture, sensor integration, communication protocols, PCB design, hardware-software integration, system debugging, low-power design, industrial applications and end-to-end project execution. The course is designed for EEE, ECE, CSE, Applied Physics, Mechatronics and relevant engineering graduates who want to build career readiness for job roles such as Embedded Systems Engineer, Firmware Engineer, IoT Engineer, Hardware Design Engineer, PCB Design Engineer, Application Engineer and Smart Systems Engineer. The training will be imparted by the Bangladesh Semiconductor Industry Association (BSIA) with funding support from SICIP.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for the occupation which comprises Units of Competency and their corresponding Elements.

OVERVIEW

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies.
- encourages the development and delivery of flexible training that suits individual and industry requirements.
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

The Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at a training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides
- Together all the parts of a Unit of Competency:
- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for the occupation were made by experts from the semiconductor, embedded systems, IoT and smart electronics industry in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competency:

- An overview of all Units of Competency for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (08 hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SEMI-SSEE-01-G	Apply OHS, ESD, Electrical and Laboratory Safety Practices	1. Identify OHS, ESD and electrical safety requirements 2. Practice safe handling of tools, devices and electronic components 3. Report hazards, faults and unsafe conditions 4. Respond to emergencies and maintain safe workplace	04
SICIP-SEMI-SSEE-02-G	Carry Out Workplace Communication, Documentation and Professional Ethics	1. Interpret workplace communication and instructions 2. Prepare technical notes, reports and documentation 3. Participate in team meetings and project reviews 4. Practice confidentiality, ethics and responsible use of equipment	04
Total Hours			8 Hrs.

Sector Specific Competencies (32 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SEMI-SSEE-01-S	Interpret Embedded Systems, Semiconductor Interfaces and Industry Applications	1. Describe embedded systems, MCUs, SoCs and semiconductor application interfaces 2. Identify industrial, IoT, automation and consumer application areas	08

Code	Unit of Competency	Elements of Competency	Duration (Hours)
		3. Interpret datasheets, errata and device selection criteria 4. Recognize power devices, sensors and interface ICs	
SICIP-SEMI-SSEE-02-S	Operate Embedded Development Environment, C Programming and Version Control	1. Set up toolchains, IDEs and cross-compilation environment 2. Apply C programming fundamentals for embedded systems 3. Use Git and project folder structure for firmware development 4. Debug basic firmware using simulator or hardware target	12
SICIP-SEMI-SSEE-03-S	Apply Basic Electronics, MCU Architecture and Hardware Resources	1. Explain MCU architecture, memory and peripherals 2. Configure GPIO, timers, PWM, ADC, DAC and interrupts 3. Apply bitwise operations and real-time control concepts 4. Use basic electronic components, breadboards and measurement tools	12
Total Hours			32 Hrs.

Occupation Specific Competencies (200 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SEMI-SSEE-01-O	Develop Embedded Firmware and MCU Applications	1. Develop bare-metal firmware for MCU platforms	28

Code	Unit of Competency	Elements of Competency	Duration (Hours)
		2. Configure GPIO, timers, PWM, interrupts and memory resources 3. Implement ADC/DAC-based data acquisition 4. Verify and document embedded applications	
SICIP-SEMI-SSEE-02-O	Integrate Sensors and Communication Interfaces	1. Select and interface analog and digital sensors 2. Implement UART, I2C, SPI and relevant communication interfaces 3. Calibrate sensors and filter sensor data 4. Develop multi-sensor data logging and display output	32
SICIP-SEMI-SSEE-03-O	Apply Hardware-Software Co-design and Semiconductor Device Selection	1. Analyze datasheets, errata and electrical characteristics 2. Select MCUs, ICs, regulators and protection devices 3. Develop firmware architecture using polling, interrupts and FSMs 4. Integrate hardware peripherals with optimized software control	32
SICIP-SEMI-SSEE-04-O	Develop PCB-based Embedded Systems	1. Prepare electronic schematics for embedded systems 2. Create PCB layout, stack-up, placement and routing 3. Apply power delivery, grounding, SI and EMC/EMI principles 4. Generate manufacturing files and prepare prototypes	36

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SEMI-SSEE-05-O	Perform System Debugging, Testing and Low-power Design	1. Use oscilloscopes, logic analyzers and JTAG/SWD for debugging 2. Decode protocols and troubleshoot hardware-firmware faults 3. Apply low-power design practices and power profiling 4. Prepare test reports and corrective action records	36
SICIP-SEMI-SSEE-06-O	Develop Industrial Smart System Application and Capstone Project	1. Plan an end-to-end smart system project 2. Develop schematic, PCB, firmware and interface integration 3. Bring up, test and demonstrate a prototype system 4. Present project evidence and demonstrate industry readiness	36
Total Hours			200 Hrs.

COMPETENCY STANDARD: SMART SYSTEMS ENGINEERING: EMBEDDED DESIGN AND SEMICONDUCTOR APPLICATION INTERFACES

The Generic Competencies

Unit of Competency: APPLY OHS, ESD, ELECTRICAL AND LABORATORY SAFETY PRACTICES	Nominal Duration: 04 hrs.	Unit Code: SICIP-SEMI- SSEE-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply OHS, ESD, electrical and laboratory safety practices in an embedded systems and electronics training/workplace environment. It specifically includes the tasks of identifying OHS, ESD and electrical safety requirements, practicing safe handling of tools, devices and electronic components, reporting hazards, faults and unsafe conditions and responding to emergencies and maintaining a safe workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS, ESD and electrical safety requirements	1.1. <u>Workplace OHS, ESD and electrical safety requirements</u> are identified according to organizational procedures. 1.2. <u>Hazards</u> associated with embedded systems and electronics laboratory environments are recognized. 1.3. <u>Personal protective equipment (PPE) and ESD protection devices</u> are identified and used appropriately. 1.4. Safe operating limits of electrical equipment and instruments are recognized and observed. 1.5. <u>Workplace safety signs, symbols and instructions</u> are interpreted correctly.
2. Practice safe handling of tools, devices and electronic components	2.1. <u>Workstations</u> are prepared according to OHS, ESD and electrical safety requirements. 2.2. <u>Tools, instruments and laboratory equipment</u> are operated following safe work procedures. 2.3. <u>Electronic components, boards and modules</u> are handled using ESD-safe practices. 2.4. Soldering, prototyping and measurement activities are carried out safely.

Elements of Competency	Performance Criteria
	2.5. Good housekeeping practices are maintained to ensure a safe working environment.
3. Report hazards, faults and unsafe conditions	3.1. Hazards, faulty equipment and unsafe conditions are identified promptly. 3.2. Workplace incidents and near-misses are reported to the concerned authority according to organizational procedures. 3.3. <u>Reporting tools and documents</u> are used to record hazards and incidents. 3.4. Corrective and preventive actions are followed according to workplace instructions. 3.5. Damaged tools and instruments are isolated, tagged and reported for repair or replacement.
4. Respond to emergencies and maintain safe workplace	4.1. <u>Emergency procedures</u> are identified according to workplace requirements. 4.2. Emergency exits, assembly points and safety equipment are located and accessed correctly. 4.3. Appropriate actions are taken during electrical, fire or other workplace emergencies. 4.4. First-aid and fire-safety instructions are applied when required. 4.5. <u>Emergency response activities</u> are carried out in accordance with workplace safety requirements.

Range of Variables

Variable	Range
	May include but not limited to:
1. Workplace OHS, ESD and electrical safety requirements	1.1. OHS Policies 1.2. ESD Procedures 1.3. Electrical Safety Rules 1.4. Safe Work Practices 1.5. Emergency Procedures 1.6. Risk Assessment
2. Hazards	2.1. Electrical Shock 2.2. Short Circuit 2.3. Electrostatic Discharge (ESD) Damage

Variable	Range May include but not limited to:
	2.4. Hot Surfaces (Soldering Iron) 2.5. Sharp Leads and Tools 2.6. Chemical Fumes from Soldering
3. Personal protective equipment (PPE) and ESD protection devices	3.1. ESD Wrist Strap 3.2. ESD Mat 3.3. ESD-safe Footwear 3.4. Safety Glasses 3.5. Heat-resistant Gloves 3.6. Fume Extractor
4. Workplace safety signs, symbols and instructions	4.1. Warning Signs 4.2. Mandatory Signs 4.3. Prohibition Signs 4.4. Emergency Exit Signs 4.5. Fire Safety Signs 4.6. Hazard Symbols
5. Workstations	5.1. ESD Workstation 5.2. Soldering Station 5.3. Computer Workstation 5.4. Test and Measurement Bench 5.5. Prototyping Workbench 5.6. Simulation Workstation
6. Tools, instruments and laboratory equipment	6.1. Soldering Station 6.2. DC Power Supply 6.3. Digital Multimeter 6.4. Oscilloscope 6.5. Logic Analyzer 6.6. Hand Tools
7. Electronic components, boards and modules	7.1. MCU Development Boards 7.2. Sensor Modules 7.3. ICs and Discrete Components 7.4. Printed Circuit Boards (PCBs) 7.5. Connectors and Cables 7.6. Display Modules

Variable	Range May include but not limited to:
8. Reporting tools and documents	8.1. Incident Report Form 8.2. Hazard Register 8.3. Logbook 8.4. Verbal Reporting 8.5. Email 8.6. Maintenance Request
9. Emergency procedures	9.1. Fire Emergency 9.2. Electrical Emergency 9.3. Medical Emergency 9.4. Evacuation Procedure 9.5. First Aid Procedure 9.6. Incident Reporting Procedure
10. Emergency response activities	10.1. Evacuation 10.2. First Aid 10.3. Firefighting 10.4. Power Isolation 10.5. Emergency Notification 10.6. Incident Documentation

Curricular Content Guide:

1. Underpinning Knowledge	1. Principles of Occupational Health and Safety (OHS), Electrostatic Discharge (ESD) protection and electrical safety in electronics laboratories. 2. Common hazards in embedded systems, soldering, prototyping and electronics work environments. 3. Proper use of Personal Protective Equipment (PPE) and ESD protection devices. 4. ESD control methods and safe handling of electronic components, boards and modules. 5. Safe operation of soldering stations, power supplies, instruments and laboratory equipment. 6. Workplace housekeeping practices and electrical safety requirements. 7. Hazard identification, incident reporting and corrective action procedures.
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	8. Emergency procedures, including fire safety, electrical hazards, evacuation routes and first aid basics.
2. Underpinning Skills	1. Applying workplace OHS, ESD and electrical safety requirements in electronics laboratory environments 2. Interpreting workplace safety signs, symbols and safety instructions 3. Preparing and maintaining safe workstations and ESD-protected work areas 4. Handling electronic components, development boards and instruments safely 5. Carrying out soldering, prototyping and measurement activities safely 6. Reporting hazards, incidents and unsafe conditions using workplace procedures 7. Following emergency response procedures for electrical, fire and workplace emergencies
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Workplace (simulated or actual) 2. Learning/operational manual 3. PPEs and ESD protection equipment 4. Firefighting equipment 5. Emergency response manual 6. Emergency Systems (Fire alarms, emergency exits, etc.) 7. First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. identified OHS, ESD and electrical safety requirements
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	2. practiced safe handling of tools, devices and electronic components 3. reported hazards, faults and unsafe conditions 4. responded to emergencies and maintained a safe workplace
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WORKPLACE COMMUNICATION, DOCUMENTATION AND PROFESSIONAL ETHICS	Nominal Duration: 04 hrs.	Unit Code: SICIP-SEMI-SSEE-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out workplace communication, documentation and professional ethics in embedded systems engineering activities. It specifically includes the tasks of interpreting workplace communication and instructions, preparing technical notes, reports and documentation, participating in team meetings and project reviews and practicing confidentiality, ethics and responsible use of equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and instructions	1.1. <u>Workplace communication methods</u> and organizational protocols are identified. 1.2.Technical instructions and work requirements are read and interpreted accurately. 1.3. <u>Professional etiquette</u> is demonstrated during workplace interactions. 1.4.Questions are asked and responded to clearly and courteously. 1.5.Team communication and collaboration are maintained according to workplace requirements.
2. Prepare technical notes, reports and documentation	2.1. <u>Technical information</u> is collected and organized according to project requirements. 2.2.Lab notes, project logs and test records are maintained accurately. 2.3. <u>Technical documents</u> are prepared using standard formats. 2.4. <u>Technical terminology and documentation standards</u> are applied correctly. 2.5.Documents are reviewed for accuracy, completeness and consistency before submission.
3. Participate in team meetings and project reviews	3.1. <u>Design data and supporting materials</u> are prepared for meetings and reviews.

Elements of Competency	Performance Criteria
	3.2.Meetings are attended punctually and own views are expressed clearly. 3.3.Feedback and review comments are recorded and analyzed. 3.4.Meeting outcomes and action items are implemented within agreed timeframes. 3.5.Professional communication and teamwork are demonstrated throughout the review process.
4. Practice confidentiality, ethics and responsible use of equipment	4.1.Confidentiality of design files and project information is maintained. 4.2.<u>Software, hardware and licenses</u> are used responsibly according to organizational policies. 4.3.Test results and project data are reported honestly and accurately. 4.4.<u>Professional ethics</u> are practiced in all workplace activities. 4.5.Equipment and shared resources are used and returned according to workplace procedures.

Range of Variables

Variable	Range
	May include but not limited to:
1. Workplace communication methods	1.1. Face-to-Face Communication 1.2. Email 1.3. Phone Calls 1.4. Video Conferencing 1.5. Instant Messaging 1.6. Meeting Notes
2. Professional etiquette	2.1. Respectful Behaviour 2.2. Professional Language 2.3. Punctuality 2.4. Active Listening 2.5. Appropriate Dress Code 2.6. Workplace Ethics
3. Technical information	3.1. Design Specifications 3.2. Project Requirements

Variable	Range May include but not limited to:
	3.3. Schematics 3.4. Test Results 3.5. Firmware Versions 3.6. Reference Documents
4. Technical documents	4.1. Lab Report 4.2. Bill of Materials (BOM) 4.3. Schematic Notes 4.4. Firmware Comments 4.5. Test Report 4.6. Project Presentation
5. Technical terminology and documentation standards	5.1. Technical Terms 5.2. Industry Standards 5.3. Naming Conventions 5.4. Document Templates 5.5. File Formats 5.6. Version Control Records
6. Design data and supporting materials	6.1. Schematics 6.2. Firmware Source Code 6.3. Presentation Slides 6.4. Technical Reports 6.5. Measurement Results 6.6. Design Specifications
7. Software, hardware and licenses	7.1. Licensed Software/IDE 7.2. Open-source Tools 7.3. Development Boards 7.4. Laboratory Instruments 7.5. Computers 7.6. Network Resources
8. Professional ethics	8.1. Confidentiality 8.2. Honesty in Test Results 8.3. Responsible Use of Tools 8.4. Respectful Teamwork 8.5. Intellectual Property Compliance

Variable	Range May include but not limited to:
	8.6. Conflict of Interest Avoidance

Curricular Content Guide:

1. Underpinning Knowledge	1. Principles of workplace communication, professional etiquette and organizational communication protocols. 2. Methods for interpreting technical instructions, work requirements and engineering documentation. 3. Standards and formats for preparing lab notes, technical reports and project documentation. 4. Use of technical terminology, documentation conventions and version control practices. 5. Principles of effective presentation, technical discussion and project review processes. 6. Methods for receiving and incorporating technical feedback into project improvements. 7. Professional ethics, confidentiality and responsible use of software, hardware and licenses. 8. Teamwork, collaboration and professional communication practices in engineering.
2. Underpinning Skills	1. Applying professional workplace communication and organizational etiquette 2. Interpreting technical instructions and work requirements accurately 3. Preparing lab notes, technical reports and project documentation using standard formats 4. Maintaining firmware versions, test records and project logs 5. Presenting project outcomes clearly during meetings and reviews 6. Incorporating feedback to improve project quality and outcomes 7. Practicing confidentiality, professional ethics and responsible use of equipment
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns

	5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Classroom or laboratory (simulated or actual workplace) 2. Computer with office tools 3. Project templates and lab notebooks 4. Learning manuals 5. Technical documents

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. interpreted workplace communication and instructions 2. prepared technical notes, reports and documentation 3. participated in team meetings and project reviews 4. practiced confidentiality, ethics and responsible use of equipment
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Sector Specific Competencies

Unit of Competency: INTERPRET EMBEDDED SYSTEMS, SEMICONDUCTOR INTERFACES AND INDUSTRY APPLICATIONS	Nominal Duration: 08 hrs.	Unit Code: SICIP-SEMI- SSEE-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to interpret embedded system architecture, MCU/SoC platforms, semiconductor application interfaces and their use in industrial, IoT, automation and consumer electronic systems. It specifically includes the tasks of describing embedded systems, MCUs, SoCs and semiconductor application interfaces, identifying industrial, IoT, automation and consumer application areas, interpreting datasheets, errata and device selection criteria and recognizing power devices, sensors and interface ICs.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Describe embedded systems, MCUs, SoCs and semiconductor application interfaces	1.1.<u>Embedded system architecture and building blocks</u> are explained according to their functional roles. 1.2.<u>MCU and SoC platforms</u> are identified and distinguished. 1.3.<u>Semiconductor application interfaces</u> are described in terms of function and use. 1.4.Hardware and software interaction in embedded products is described. 1.5.Typical embedded product development flow from requirement to deployment is interpreted.
2. Identify industrial, IoT, automation and consumer application areas	2.1.<u>Application areas</u> of embedded and smart systems are identified. 2.2.Use cases for monitoring, control and data acquisition systems are interpreted. 2.3.<u>Application constraints</u> such as cost, power and reliability are identified. 2.4.Job roles and career pathways in embedded systems and semiconductor application engineering are recognized. 2.5.Industry trends and employment opportunities are interpreted.

Elements of Competency	Performance Criteria
3. Interpret datasheets, errata and device selection criteria	3.1. <u>Datasheet sections</u> are located and interpreted correctly. 3.2. Electrical characteristics, absolute maximum ratings and package information are read accurately. 3.3. <u>Errata sheets and application notes</u> are considered during design decisions. 3.4. <u>Device selection criteria</u> are identified according to application requirements. 3.5. Device comparison summaries are prepared based on datasheet parameters.
4. Recognize power devices, sensors and interface ICs	4.1. <u>Power devices and regulators</u> are identified according to their functions. 4.2. <u>Sensor types</u> and their interface requirements are identified. 4.3. <u>Interface and protection ICs</u> are recognized according to application use. 4.4. Package types and mounting technologies are distinguished. 4.5. Typical application circuits of common semiconductor devices are interpreted.

Range of Variables

Variable	Range
	May include but not limited to:
1. Embedded system architecture and building blocks	1.1. Processor Core 1.2. Memory (Flash, SRAM, EEPROM) 1.3. Peripherals 1.4. Clock and Reset Circuits 1.5. Power Supply 1.6. Input/Output Interfaces
2. MCU and SoC platforms	2.1. Arduino (AVR) 2.2. ESP32 2.3. STM32 2.4. Raspberry Pi Pico (RP2040) 2.5. PIC 2.6. Equivalent MCU/SoC Platforms

Variable	Range May include but not limited to:
3. Semiconductor application interfaces	3.1. GPIO 3.2. UART 3.3. I2C 3.4. SPI 3.5. ADC/DAC 3.6. PWM
4. Application areas	4.1. Industrial Monitoring 4.2. Automation and Control 4.3. IoT Nodes 4.4. Consumer Electronics 4.5. Data Logging 4.6. Smart Agriculture and Energy
5. Application constraints	5.1. Cost 5.2. Power Consumption 5.3. Reliability 5.4. Performance 5.5. Size/Form Factor 5.6. Environmental Conditions
6. Datasheet sections	6.1. Feature Summary 6.2. Pin Configuration 6.3. Electrical Characteristics 6.4. Absolute Maximum Ratings 6.5. Typical Application Circuit 6.6. Package Information
7. Errata sheets and application notes	7.1. Silicon Errata 7.2. Design Guidelines 7.3. Reference Designs 7.4. Layout Recommendations 7.5. Migration Guides 7.6. Safety Notes
8. Device selection criteria	8.1. Operating Voltage 8.2. Current Consumption 8.3. Speed/Frequency

Variable	Range May include but not limited to:
	8.4. Memory Size 8.5. Peripheral Set 8.6. Availability and Cost
9. Power devices and regulators	9.1. LDO Regulator 9.2. Switching Regulator (Buck/Boost) 9.3. MOSFET 9.4. Diode 9.5. Power Management IC 9.6. Battery Charger IC
10. Sensor types	10.1. Temperature Sensor 10.2. Humidity Sensor 10.3. Pressure Sensor 10.4. Motion/Accelerometer Sensor 10.5. Light Sensor 10.6. Gas Sensor
11. Interface and protection ICs	11.1. Level Shifter 11.2. Line Driver/Transceiver 11.3. ESD Protection Device 11.4. Optocoupler/Isolator 11.5. Display Driver 11.6. Motor Driver

Curricular Content Guide:

1. Underpinning Knowledge	1. Embedded systems fundamentals, architecture and building blocks. 2. MCU/SoC architecture overview and platform comparison. 3. Semiconductor devices and application interfaces (GPIO, UART, I2C, SPI, ADC/DAC, PWM). 4. Industrial, IoT, automation and consumer application contexts and constraints. 5. Datasheet, errata and application note structure and interpretation. 6. Device selection methodology based on electrical and application parameters.
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	7. Job roles, career pathways and industry trends in the embedded and semiconductor sector.
2. Underpinning Skills	1. Identifying embedded platforms and their building blocks 2. Reading datasheets, errata sheets and application notes 3. Interpreting electrical characteristics and package information 4. Selecting devices based on application requirements 5. Explaining smart system applications and their constraints 6. Preparing device comparison summaries
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Computer laboratory 2. Development boards and datasheets 3. Sensor and interface IC samples 4. Internet access and learning materials 5. Learning manuals

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. described embedded systems, MCUs, SoCs and semiconductor application interfaces 2. identified industrial, IoT, automation and consumer application areas 3. interpreted datasheets, errata and device selection criteria 4. recognized power devices, sensors and interface ICs
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration

	3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE EMBEDDED DEVELOPMENT ENVIRONMENT, C PROGRAMMING AND VERSION CONTROL	Nominal Duration: 12 hrs.	Unit Code: SICIP-SEMI-SSEE-02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate the embedded development environment, apply C programming for embedded systems and use version control for firmware development. It specifically includes the tasks of setting up toolchains, IDEs and cross-compilation environment, applying C programming fundamentals for embedded systems, using Git and project folder structure for firmware development and debugging basic firmware using simulator or hardware target.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Set up toolchains, IDEs and cross-compilation environment	1.1. <u>Development tools and IDEs</u> are installed and configured according to platform requirements. 1.2. <u>Toolchain components</u> are identified and set up for cross-compilation. 1.3.Board support packages and device drivers are installed correctly. 1.4.Project build settings and target configurations are established. 1.5.Toolchain installation is verified by building and running a sample program.
2. Apply C programming fundamentals for embedded systems	2.1. <u>C language constructs</u> are applied to write structured programs. 2.2. <u>Data types, operators and control structures</u> are used correctly. 2.3.Functions, arrays, pointers and structures are implemented in embedded programs. 2.4. <u>Bitwise operations</u> are applied for register-level programming. 2.5.Programs are compiled, debugged and corrected for syntax and logic errors.
3. Use Git and project folder structure for firmware development	3.1. <u>Version control operations</u> are performed using Git. 3.2.Repositories are created, cloned and managed according to project requirements.

Elements of Competency	Performance Criteria
	3.3.<u>Project folder structures</u> are organized following standard conventions. 3.4.Commit messages and change histories are maintained meaningfully. 3.5.Code is pushed, pulled and merged with remote repositories correctly.
4. Debug basic firmware using simulator or hardware target	4.1.<u>Debugging tools and methods</u> are selected according to the development environment. 4.2.Breakpoints, watch variables and step execution are used to trace program flow. 4.3.Firmware is downloaded/flashed to the target board successfully. 4.4.Program outputs are verified against expected results. 4.5.<u>Common firmware errors</u> are identified and corrected.

Range of Variables

Variable	Range
	May include but not limited to:
1. Development tools and IDEs	1.1. Arduino IDE 1.2. STM32CubeIDE 1.3. PlatformIO 1.4. VS Code 1.5. MPLAB X 1.6. Keil MDK
2. Toolchain components	2.1. Cross-compiler (GCC-ARM) 2.2. Assembler 2.3. Linker 2.4. Debugger (GDB/OpenOCD) 2.5. Flash Programmer 2.6. Makefile/Build System
3. C language constructs	3.1. Variables and Constants 3.2. Control Statements 3.3. Loops 3.4. Functions 3.5. Preprocessor Directives

Variable	Range May include but not limited to:
	3.6. Header Files
4. Data types, operators and control structures	4.1. int/char/float 4.2. Arrays 4.3. Pointers 4.4. Structures and Unions 4.5. Arithmetic/Logical Operators 4.6. if-else/switch/for/while
5. Bitwise operations	5.1. AND 5.2. OR 5.3. XOR 5.4. NOT 5.5. Shift Left/Right 5.6. Bit Set/Clear/Toggle/Mask
6. Version control operations	6.1. init 6.2. clone 6.3. add/commit 6.4. push/pull 6.5. branch/merge 6.6. status/log
7. Project folder structures	7.1. Source (src) 7.2. Include (inc) 7.3. Drivers 7.4. Documentation (docs) 7.5. Build Output 7.6. README and License Files
8. Debugging tools and methods	8.1. Simulator 8.2. On-board Debugger 8.3. JTAG/SWD Probe 8.4. Serial Print Debugging 8.5. LED Indication 8.6. IDE Debugger
9. Common firmware errors	9.1. Syntax Errors 9.2. Logic Errors

Variable	Range May include but not limited to:
	9.3. Configuration Errors 9.4. Memory Overflow 9.5. Infinite Loops 9.6. Incorrect Register Settings

Curricular Content Guide:

1. Underpinning Knowledge	1. Embedded development environments, toolchains and cross-compilation concepts. 2. C programming fundamentals for embedded systems. 3. Data types, pointers, structures and bitwise operations for register-level programming. 4. Version control concepts and Git workflow for firmware development. 5. Standard firmware project organization and folder conventions. 6. Debugging concepts using simulators and hardware targets. 7. Common firmware errors and correction techniques.
2. Underpinning Skills	1. Installing and configuring IDEs, toolchains and board support packages 2. Writing structured C programs for embedded targets 3. Applying bitwise operations for register-level programming 4. Using Git for version control and collaboration 5. Organizing firmware project folders following standard conventions 6. Debugging firmware using simulator or hardware target
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace

4. Resource Implications	The following resources must be provided: 1. Computer laboratory with development software 2. MCU development boards and programming cables 3. Internet access 4. Learning manuals and sample codes 5. Git and repository access
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. set up toolchains, IDEs and cross-compilation environment 2. applied C programming fundamentals for embedded systems 3. used Git and project folder structure for firmware development 4. debugged basic firmware using simulator or hardware target
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY BASIC ELECTRONICS, MCU ARCHITECTURE AND HARDWARE RESOURCES	Nominal Duration: 12 hrs.	Unit Code: SICIP-SEMI-SSEE-03-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply basic electronics, interpret MCU architecture and configure hardware resources for embedded applications. It specifically includes the tasks of explaining MCU architecture, memory and peripherals, configuring GPIO, timers, PWM, ADC, DAC and interrupts, applying bitwise operations and real-time control concepts and using basic electronic components, breadboards and measurement tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain MCU architecture, memory and peripherals	1.1. <u>MCU architecture blocks</u> are explained according to their functional roles. 1.2. <u>Memory types</u> and memory maps are interpreted correctly. 1.3.Peripheral functions and register-level control concepts are described. 1.4.Clock sources, reset circuits and power supply requirements are identified. 1.5.Bus structures and data flow within the MCU are interpreted.
2. Configure GPIO, timers, PWM, ADC, DAC and interrupts	2.1. <u>GPIO pins</u> are configured as inputs and outputs using registers or libraries. 2.2. <u>Timers and PWM channels</u> are configured for timing and control applications. 2.3. <u>ADC and DAC resources</u> are configured for analog signal handling. 2.4. <u>Interrupt sources</u> are configured and interrupt service routines are implemented. 2.5.Peripheral operations are verified using test programs and measurements.
3. Apply bitwise operations and real-time control concepts	3.1.Bitwise operations are applied to set, clear, toggle and mask register bits.

Elements of Competency	Performance Criteria
	3.2.<u>Real-time control concepts</u> are applied in program design. 3.3.Polling and interrupt-driven approaches are distinguished and applied appropriately. 3.4.Timing requirements and delays are implemented accurately. 3.5.Program responsiveness is verified against real-time requirements.
4. Use basic electronic components, breadboards and measurement tools	4.1.<u>Basic electronic components</u> are identified and their functions explained. 4.2.Component values and ratings are read and verified correctly. 4.3.<u>Breadboard circuits</u> are constructed according to circuit diagrams. 4.4.<u>Measurement tools</u> are used to measure voltage, current, resistance and signals. 4.5.Circuit operation is tested and faults are identified and corrected.

Range of Variables

Variable	Range May include but not limited to:
1. MCU architecture blocks	1.1. CPU Core 1.2. Flash Memory 1.3. SRAM 1.4. Peripherals 1.5. Clock System 1.6. Interrupt Controller
2. Memory types	2.1. Flash 2.2. SRAM 2.3. EEPROM 2.4. ROM 2.5. Register Memory 2.6. External Memory
3. GPIO pins	3.1. Digital Input 3.2. Digital Output

Variable	Range May include but not limited to:
	3.3. Pull-up/Pull-down 3.4. Open-drain 3.5. Alternate Function 3.6. Analog Mode
4. Timers and PWM channels	4.1. Basic Timer 4.2. General-purpose Timer 4.3. PWM Output 4.4. Input Capture 4.5. Output Compare 4.6. Watchdog Timer
5. ADC and DAC resources	5.1. ADC Channels 5.2. Resolution (8/10/12-bit) 5.3. Sampling Rate 5.4. Reference Voltage 5.5. DAC Output 5.6. Analog Multiplexer
6. Interrupt sources	6.1. External Interrupt 6.2. Timer Interrupt 6.3. UART Interrupt 6.4. ADC Interrupt 6.5. SysTick 6.6. NVIC Priority
7. Real-time control concepts	7.1. Polling 7.2. Interrupts 7.3. Timing and Delays 7.4. Task Scheduling 7.5. State Machines 7.6. Response Time
8. Basic electronic components	8.1. Resistor 8.2. Capacitor 8.3. Diode 8.4. LED 8.5. Transistor/MOSFET

Variable	Range May include but not limited to:
	8.6. Push Button and Switch
9. Breadboard circuits	9.1. LED Drive Circuit 9.2. Switch Input Circuit 9.3. Sensor Interface Circuit 9.4. Voltage Divider 9.5. Transistor Switch Circuit 9.6. RC Filter
10. Measurement tools	10.1. Digital Multimeter 10.2. Oscilloscope 10.3. Logic Analyzer 10.4. DC Power Supply 10.5. Function Generator 10.6. Continuity Tester

Curricular Content Guide:

1. Underpinning Knowledge	1. Basic electronics: components, circuit laws and signal fundamentals. 2. MCU architecture, memory organization and bus structure. 3. GPIO, timer, PWM, ADC, DAC and interrupt operation. 4. Register-level programming and bitwise control concepts. 5. Polling versus interrupt-driven design and real-time control concepts. 6. Breadboard prototyping and circuit construction practices. 7. Use of measurement instruments for circuit testing and verification.
2. Underpinning Skills	1. Explaining MCU architecture, memory and peripherals 2. Configuring GPIO, timers, PWM, ADC, DAC and interrupts 3. Applying bitwise operations for register control 4. Implementing polling and interrupt-driven programs 5. Constructing breadboard circuits from circuit diagrams 6. Measuring circuit parameters with instruments and verifying operation
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities

	3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Electronics laboratory 2. MCU development boards 3. Breadboards and component kits 4. Measurement instruments (multimeter, oscilloscope, logic analyzer, power supply) 5. Learning manuals and datasheets

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. explained MCU architecture, memory and peripherals 2. configured GPIO, timers, PWM, ADC, DAC and interrupts 3. applied bitwise operations and real-time control concepts 4. used basic electronic components, breadboards and measurement tools
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Occupation Specific Competencies

Unit of Competency: DEVELOP EMBEDDED FIRMWARE AND MCU APPLICATIONS	Nominal Duration: 28 hrs.	Unit Code: SICIP-SEMI-SSEE-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to develop embedded firmware and MCU applications. It specifically includes the tasks of developing bare-metal firmware for MCU platforms, configuring GPIO, timers, PWM, interrupts and memory resources, implementing ADC/DAC-based data acquisition and verifying and documenting embedded applications.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Develop bare-metal firmware for MCU platforms	1.1. <u>Firmware requirements</u> are interpreted according to application specifications. 1.2.Firmware architecture and program flow are planned before coding. 1.3. <u>Bare-metal firmware</u> is written using register-level or HAL-based programming. 1.4.Startup, initialization and main loop structures are implemented correctly. 1.5.Firmware is compiled, flashed and executed successfully on the target platform.
2. Configure GPIO, timers, PWM, interrupts and memory resources	2.1.GPIO pins are configured and controlled according to application requirements. 2.2. <u>Timer and PWM configurations</u> are implemented for timing and control functions. 2.3. <u>Interrupt-driven routines</u> are implemented with appropriate priorities. 2.4. <u>Memory resources</u> are allocated and managed efficiently. 2.5.Peripheral operation is verified through test routines and measurements.
3. Implement ADC/DAC-based data acquisition	3.1. <u>ADC configurations</u> are implemented for analog signal acquisition. 3.2.Sampling rate, resolution and reference settings are selected appropriately.

Elements of Competency	Performance Criteria
	3.3.Acquired data is scaled, converted and processed correctly. 3.4.<u>DAC and analog output functions</u> are implemented where required. 3.5.Data acquisition accuracy is verified against reference measurements.
4. Verify and document embedded applications	4.1.<u>Test procedures</u> are prepared according to application requirements. 4.2.Firmware functionality is tested against specifications. 4.3.Defects are identified, logged and corrected. 4.4.<u>Firmware documentation</u> is prepared using standard formats. 4.5.Verified firmware and documentation are stored under version control.

Range of Variables

Variable	Range
	May include but not limited to:
1. Firmware requirements	1.1. Functional Requirements 1.2. Timing Requirements 1.3. I/O Requirements 1.4. Memory Constraints 1.5. Power Constraints 1.6. Interface Requirements
2. Bare-metal firmware	2.1. Register-level Code 2.2. HAL/Driver-based Code 2.3. Startup Code 2.4. Interrupt Service Routines 2.5. Main Loop 2.6. Device Drivers
3. Timer and PWM configurations	3.1. Periodic Interrupt 3.2. Delay Generation 3.3. PWM Duty Cycle Control 3.4. Frequency Setting 3.5. Input Capture

Variable	Range May include but not limited to:
	3.6. Servo/Motor Control
4. Interrupt-driven routines	4.1. External Interrupt 4.2. Timer Interrupt 4.3. ADC Interrupt 4.4. UART Interrupt 4.5. Priority Configuration 4.6. ISR Design
5. Memory resources	5.1. Flash 5.2. SRAM 5.3. Stack 5.4. Heap 5.5. Constant Data 5.6. Buffers
6. ADC configurations	6.1. Single Conversion 6.2. Continuous Conversion 6.3. Multi-channel Scanning 6.4. Interrupt-based Reading 6.5. Averaging/Filtering 6.6. Calibration
7. DAC and analog output functions	7.1. DAC Output 7.2. PWM-based Analog Output 7.3. Waveform Generation 7.4. Signal Level Adjustment 7.5. Audio Tone Output 7.6. Reference Output
8. Test procedures	8.1. Functional Test 8.2. I/O Test 8.3. Timing Test 8.4. Boundary Test 8.5. Stress Test 8.6. Regression Test
9. Firmware documentation	9.1. Source Code Comments 9.2. README File

Variable	Range May include but not limited to:
	9.3. Flowchart 9.4. Configuration Description 9.5. Test Report 9.6. Version History

Curricular Content Guide:

1. Underpinning Knowledge	1. Firmware development flow and architecture design. 2. Register-level and HAL-based programming techniques. 3. GPIO, timer, PWM and interrupt programming for MCU applications. 4. Memory organization and resource management in embedded systems. 5. ADC/DAC principles and data acquisition techniques. 6. Firmware testing and verification methods. 7. Firmware documentation and version control practices.
2. Underpinning Skills	1. Developing bare-metal firmware for MCU platforms 2. Configuring peripherals and interrupt-driven routines 3. Implementing ADC/DAC-based data acquisition 4. Processing and scaling acquired data 5. Testing firmware against application specifications 6. Preparing firmware documentation and maintaining versions
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Computer laboratory with IDE and toolchains 2. MCU development boards 3. Sensors and I/O modules

	4. Measurement instruments 5. Learning manuals and datasheets
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. developed bare-metal firmware for MCU platforms 2. configured GPIO, timers, PWM, interrupts and memory resources 3. implemented ADC/DAC-based data acquisition 4. verified and documented embedded applications
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: INTEGRATE SENSORS AND COMMUNICATION INTERFACES	Nominal Duration: 32 hrs.	Unit Code: SICIP-SEMI-SSEE-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to integrate sensors and communication interfaces into embedded applications. It specifically includes the tasks of selecting and interfacing analog and digital sensors, implementing UART, I2C, SPI and relevant communication interfaces, calibrating sensors and filtering sensor data and developing multi-sensor data logging and display output.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select and interface analog and digital sensors	1.1. <u>Sensor requirements</u> are interpreted according to application specifications. 1.2. <u>Analog and digital sensors</u> are selected based on datasheet parameters. 1.3. Sensor interfacing circuits are designed and connected correctly. 1.4. Sensor signals are acquired using appropriate MCU resources. 1.5. Sensor operation is verified against expected outputs.
2. Implement UART, I2C, SPI and relevant communication interfaces	2.1. <u>Communication interfaces</u> are configured according to protocol specifications. 2.2. UART communication is implemented for data transmission and reception. 2.3. I2C-based devices are addressed, read and written correctly. 2.4. SPI-based devices are interfaced with correct modes and timing. 2.5. <u>Communication errors</u> are detected and handled appropriately.
3. Calibrate sensors and filter sensor data	3.1. <u>Calibration methods</u> are applied according to sensor characteristics. 3.2. Offset and gain corrections are implemented in firmware. 3.3. <u>Filtering techniques</u> are applied to reduce noise in sensor data.

Elements of Competency	Performance Criteria
	3.4. Calibrated readings are validated against reference instruments. 3.5. Calibration records are documented for traceability.
4. Develop multi-sensor data logging and display output	4.1. Multiple sensors are integrated into a single embedded application. 4.2. <u>Data logging methods</u> are implemented for storage and retrieval. 4.3. <u>Display interfaces</u> are implemented to present sensor data. 4.4. Timestamping and data formatting are applied correctly. 4.5. Complete multi-sensor system operation is tested and documented.

Range of Variables

Variable	Range May include but not limited to:
1. Sensor requirements	1.1. Measurement Range 1.2. Accuracy 1.3. Resolution 1.4. Response Time 1.5. Interface Type 1.6. Operating Conditions
2. Analog and digital sensors	2.1. Temperature Sensor (LM35/DS18B20) 2.2. Humidity Sensor (DHT11/DHT22) 2.3. Pressure Sensor (BMP180/BMP280) 2.4. Accelerometer/IMU (ADXL345/MPU6050) 2.5. Light Sensor (LDR/BH1750) 2.6. Gas Sensor (MQ Series)
3. Communication interfaces	3.1. UART/USART 3.2. I2C 3.3. SPI 3.4. 1-Wire 3.5. RS-485 3.6. CAN (Introductory)

Variable	Range May include but not limited to:
4. Communication errors	4.1. Framing Error 4.2. Parity Error 4.3. NACK/No Acknowledgement 4.4. Bus Contention 4.5. Timeout 4.6. Data Corruption
5. Calibration methods	5.1. Zero/Offset Calibration 5.2. Gain/Span Calibration 5.3. Two-point Calibration 5.4. Multi-point Calibration 5.5. Reference Comparison 5.6. Look-up Table Correction
6. Filtering techniques	6.1. Moving Average 6.2. Median Filter 6.3. Low-pass Filter 6.4. Exponential Smoothing 6.5. Debouncing 6.6. Outlier Rejection
7. Data logging methods	7.1. EEPROM Logging 7.2. SD Card Logging 7.3. Serial Logging to PC 7.4. Flash Memory Storage 7.5. Circular Buffer 7.6. Timestamped Records
8. Display interfaces	8.1. Character LCD (16x2) 8.2. OLED Display (SSD1306) 8.3. Seven-segment Display 8.4. TFT Display 8.5. Serial Terminal 8.6. LED Indicators

Curricular Content Guide:

1. Underpinning Knowledge	1. Sensor principles, characteristics and selection criteria. 2. Analog and digital sensor interfacing techniques. 3. UART, I2C and SPI protocol operation, configuration and timing. 4. Communication error detection and handling techniques. 5. Sensor calibration methods and data filtering techniques. 6. Data logging and storage methods for embedded systems. 7. Display interfacing and data presentation techniques.
2. Underpinning Skills	1. Selecting sensors from datasheet parameters 2. Designing and connecting sensor interface circuits 3. Implementing UART, I2C and SPI communication 4. Calibrating sensors and filtering sensor data 5. Developing multi-sensor data logging applications 6. Implementing display outputs and validating system operation
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Electronics laboratory 2. MCU development boards 3. Sensor modules (analog and digital) 4. Display modules 5. Measurement instruments 6. Learning manuals and datasheets

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. selected and interfaced analog and digital sensors
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	2. implemented UART, I2C, SPI and relevant communication interfaces 3. calibrated sensors and filtered sensor data 4. developed multi-sensor data logging and display output
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY HARDWARE-SOFTWARE CO-DESIGN AND SEMICONDUCTOR DEVICE SELECTION	Nominal Duration: 32 hrs.	Unit Code: SICIP-SEMI-SSEE-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply hardware-software co-design and select semiconductor devices for embedded applications. It specifically includes the tasks of analyzing datasheets, errata and electrical characteristics, selecting MCUs, ICs, regulators and protection devices, developing firmware architecture using polling, interrupts and FSMs and integrating hardware peripherals with optimized software control.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Analyze datasheets, errata and electrical characteristics	1.1. <u>Datasheet parameters</u> are analyzed for design suitability. 1.2.Absolute maximum ratings and recommended operating conditions are interpreted correctly. 1.3. <u>Electrical characteristics</u> are compared against application requirements. 1.4.Errata sheets and application notes are reviewed for design constraints. 1.5.Analysis results are documented to support design decisions.
2. Select MCUs, ICs, regulators and protection devices	2.1. <u>MCU selection criteria</u> are applied according to application requirements. 2.2. <u>Voltage regulators and power devices</u> are selected based on load and efficiency requirements. 2.3. <u>Level shifting and isolation devices</u> are selected for interface compatibility. 2.4. <u>Protection devices</u> are selected for ESD, overvoltage and overcurrent protection. 2.5.Selected devices are justified through comparison and documentation.
3. Develop firmware architecture using polling, interrupts and FSMs	3.1. <u>Firmware architecture approaches</u> are selected according to system requirements. 3.2.Polling-based program structures are implemented where appropriate.

Elements of Competency	Performance Criteria
	3.3.Interrupt-driven structures are implemented for time-critical events. 3.4.Finite state machines (FSMs) are designed and implemented for system behavior. 3.5.Firmware architecture is documented using flowcharts or state diagrams.
4. Integrate hardware peripherals with optimized software control	4.1.Hardware peripherals are integrated according to interface specifications. 4.2.Software control techniques are optimized for performance and resource usage. 4.3.Hardware-software timing and synchronization issues are resolved. 4.4.Integration tests are performed to verify combined operation. 4.5.Integration results and optimizations are documented.

Range of Variables

Variable	Range
	May include but not limited to:
1. Datasheet parameters	1.1. Operating Voltage 1.2. Current Consumption 1.3. Timing Characteristics 1.4. Thermal Characteristics 1.5. Pin Functions 1.6. Package Details
2. Electrical characteristics	2.1. Input/Output Voltage Levels 2.2. Drive Current 2.3. Leakage Current 2.4. Frequency Response 2.5. Power Dissipation 2.6. Temperature Range
3. MCU selection criteria	3.1. Core and Speed 3.2. Memory Size 3.3. Peripheral Set 3.4. Power Consumption

Variable	Range May include but not limited to:
	3.5. Package and Pin Count 3.6. Cost and Availability
4. Voltage regulators and power devices	4.1. LDO Regulator 4.2. Buck Converter 4.3. Boost Converter 4.4. MOSFET 4.5. Power Management IC 4.6. Battery Management IC
5. Level shifting and isolation devices	5.1. Level Shifter IC 5.2. Voltage Divider 5.3. Optocoupler 5.4. Digital Isolator 5.5. Bus Transceiver 5.6. Relay
6. Protection devices	6.1. TVS Diode 6.2. Zener Diode 6.3. Fuse/PTC 6.4. ESD Protection Array 6.5. Reverse Polarity Protection 6.6. Snubber Circuit
7. Firmware architecture approaches	7.1. Super Loop 7.2. Polling 7.3. Interrupt-driven 7.4. Event-driven 7.5. FSM-based 7.6. Cooperative Scheduling
8. Finite state machines (FSMs)	8.1. State Definition 8.2. Transition Conditions 8.3. State Diagram 8.4. Switch-case Implementation 8.5. Table-driven FSM 8.6. Hierarchical FSM

Variable	Range May include but not limited to:
9. Software control techniques	9.1. Code Optimization 9.2. Efficient ISR Design 9.3. Buffering 9.4. Timing Management 9.5. Resource Sharing 9.6. Low-latency Response
10. Integration tests	10.1. Interface Test 10.2. Timing Test 10.3. Load Test 10.4. Functional Integration Test 10.5. Fault Injection Test 10.6. Regression Test

Curricular Content Guide:

1. Underpinning Knowledge	1. Datasheet, errata and electrical characteristic analysis techniques. 2. Semiconductor device selection methodology and trade-offs. 3. Power regulation, level shifting, isolation and circuit protection concepts. 4. Hardware-software co-design principles. 5. Polling, interrupt-driven and FSM-based firmware architectures. 6. Hardware-software integration and optimization techniques. 7. Design documentation and justification methods.
2. Underpinning Skills	1. Analyzing datasheets, errata and electrical characteristics 2. Selecting MCUs, regulators, interface and protection devices 3. Designing firmware architectures using polling, interrupts and FSMs 4. Implementing state machine-based system behavior 5. Integrating peripherals with optimized software control 6. Documenting design decisions and integration results
3. Underpinning Attitudes	1. Commitment to occupational health and safety

	- Promptness in carrying out activities - Sincerity and honesty to duties - Environmental concerns - Eagerness to learn - Tidiness and timeliness - Respect for rights of peers and seniors in the workplace - Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: - Computer laboratory - Development boards and device samples - Datasheets and application notes - Simulation and design tools - Measurement instruments - Learning manuals

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: - analyzed datasheets, errata and electrical characteristics - selected MCUs, ICs, regulators and protection devices - developed firmware architecture using polling, interrupts and FSMs - integrated hardware peripherals with optimized software control
2. Methods of Assessment	Methods of assessment should include: - Written test - Practical demonstration - Oral questioning - Portfolio (optional)
3. Context of Assessment	- Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. - Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: DEVELOP PCB-BASED EMBEDDED SYSTEMS	Nominal Duration: 36 hrs.	Unit Code: SICIP-SEMI-SSEE-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to develop PCB-based embedded systems. It specifically includes the tasks of preparing electronic schematics for embedded systems, creating PCB layout, stack-up, placement and routing, applying power delivery, grounding, SI and EMC/EMI principles and generating manufacturing files and preparing prototypes.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare electronic schematics for embedded systems	1.1. <u>Schematic design tools</u> are used according to project requirements. 1.2. Circuit blocks are drawn using standard symbols and conventions. 1.3. <u>Component libraries and footprints</u> are selected or created correctly. 1.4. Net names, reference designators and component values are annotated correctly. 1.5. <u>Schematic verification checks</u> are performed and errors are corrected.
2. Create PCB layout, stack-up, placement and routing	2.1. <u>PCB stack-up and design rules</u> are defined according to design requirements. 2.2. Components are placed considering signal flow, thermal and mechanical constraints. 2.3. <u>Routing techniques</u> are applied for signal, power and ground connections. 2.4. Design rule checks (DRC) are performed and violations are corrected. 2.5. Layout is reviewed against the schematic and design guidelines.
3. Apply power delivery, grounding, SI and EMC/EMI principles	3.1. <u>Power delivery network components</u> are implemented in the layout. 3.2. <u>Grounding techniques</u> are applied to minimize noise and interference.

Elements of Competency	Performance Criteria
	3.3. Decoupling and bulk capacitors are placed effectively near power pins. 3.4. <u>Signal integrity and EMC/EMI practices</u> are applied to critical signals. 3.5. Layout is verified against power delivery and EMC design guidelines.
4. Generate manufacturing files and prepare prototypes	4.1. <u>Manufacturing output files</u> are generated according to fabrication requirements. 4.2. Bill of Materials (BOM) and assembly documents are prepared. 4.3. <u>DFM/DFA checks</u> are performed before design release. 4.4. Prototype boards are assembled and inspected. 4.5. Prototype bring-up tests are performed and results are documented.

Range of Variables

Variable	Range
	May include but not limited to:
1. Schematic design tools	1.1. KiCad 1.2. Eagle 1.3. Altium Designer 1.4. EasyEDA 1.5. OrCAD 1.6. Equivalent EDA Tools
2. Component libraries and footprints	2.1. Standard Symbol Library 2.2. Footprint Library 2.3. 3D Models 2.4. Custom Symbols 2.5. Custom Footprints 2.6. Manufacturer Libraries
3. Schematic verification checks	3.1. Electrical Rule Check (ERC) 3.2. Net Connectivity Check 3.3. Reference Designator Check 3.4. Value Check 3.5. Power Connection Check

Variable	Range May include but not limited to:
	3.6. Annotation Check
4. PCB stack-up and design rules	4.1. Single Layer 4.2. Double Layer 4.3. Four Layer 4.4. Trace Width/Clearance 4.5. Via Sizes 4.6. Copper Pour Rules
5. Routing techniques	5.1. Signal Routing 5.2. Power Routing 5.3. Ground Pour 5.4. Differential Pairs (Introductory) 5.5. Length Matching (Introductory) 5.6. Via Stitching
6. Power delivery network components	6.1. Decoupling Capacitors 6.2. Bulk Capacitors 6.3. Power Planes 6.4. Ferrite Beads 6.5. Voltage Regulators 6.6. Power Traces
7. Grounding techniques	7.1. Ground Plane 7.2. Star Grounding 7.3. Ground Pour 7.4. Analog-Digital Ground Separation 7.5. Ground Stitching Vias 7.6. Chassis Grounding
8. Signal integrity and EMC/EMI practices	8.1. Short Return Paths 8.2. Controlled Trace Routing 8.3. Crosstalk Reduction 8.4. Shielding 8.5. Filtering 8.6. Clock Signal Care
9. Manufacturing output files	9.1. Gerber Files 9.2. Drill Files (Excellon)

Variable	Range May include but not limited to:
	9.3. Pick-and-Place File 9.4. BOM 9.5. Assembly Drawing 9.6. PCB Fabrication Notes
10. DFM/DFA checks	10.1. Minimum Trace/Space 10.2. Drill Sizes 10.3. Solder Mask Clearance 10.4. Silkscreen Readability 10.5. Component Spacing 10.6. Panelization Considerations

Curricular Content Guide:

1. Underpinning Knowledge	1. Schematic capture standards, symbols and annotation conventions. 2. PCB stack-up, design rules and layout methodology. 3. Component placement and routing techniques for embedded boards. 4. Power delivery network design and grounding principles. 5. Signal integrity and EMC/EMI fundamentals for PCB design. 6. DFM/DFA considerations and manufacturing file requirements. 7. Prototype assembly, inspection and bring-up practices.
2. Underpinning Skills	1. Preparing schematics using EDA tools 2. Managing component libraries and footprints 3. Creating PCB layouts with correct placement and routing 4. Applying PDN, grounding and EMC/EMI practices 5. Performing ERC/DRC and DFM/DFA checks 6. Generating Gerber, drill, BOM and assembly files 7. Assembling and testing prototype boards
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn

	6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Computer laboratory with EDA software 2. PCB prototyping and soldering facilities 3. Component inventory 4. Measurement instruments 5. Learning manuals and design guidelines

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. prepared electronic schematics for embedded systems 2. created PCB layout, stack-up, placement and routing 3. applied power delivery, grounding, SI and EMC/EMI principles 4. generated manufacturing files and prepared prototypes
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM SYSTEM DEBUGGING, TESTING AND LOW-POWER DESIGN	Nominal Duration: 36 hrs.	Unit Code: SICIP-SEMI-SSEE-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform system debugging, testing and low-power design of embedded systems. It specifically includes the tasks of using oscilloscopes, logic analyzers and JTAG/SWD for debugging, decoding protocols and troubleshooting hardware-firmware faults, applying low-power design practices and power profiling and preparing test reports and corrective action records.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Use oscilloscopes, logic analyzers and JTAG/SWD for debugging	1.1. <u>Debugging instruments</u> are set up according to measurement requirements. 1.2.Oscilloscope measurements are performed on analog and digital signals. 1.3. <u>Logic analyzer captures</u> are configured and interpreted correctly. 1.4. <u>JTAG/SWD debugging sessions</u> are established with the target MCU. 1.5.Measurement results are recorded and interpreted correctly.
2. Decode protocols and troubleshoot hardware-firmware faults	2.1. <u>Protocol decoding functions</u> are applied to communication signals. 2.2.Communication timing and data integrity are analyzed. 2.3. <u>Fault symptoms</u> are identified and localized systematically. 2.4.Hardware and firmware root causes are distinguished and diagnosed. 2.5. <u>Troubleshooting techniques</u> are applied and fixes are verified.
3. Apply low-power design practices and power profiling	3.1. <u>Low-power modes</u> of the MCU are configured according to application requirements. 3.2. <u>Wake-up sources</u> are configured and verified. 3.3.Peripheral and clock management techniques are applied to reduce power consumption.

Elements of Competency	Performance Criteria
	3.4.<u>Power profiling measurements</u> are performed on the system. 3.5.Power optimization results are analyzed and documented.
4. Prepare test reports and corrective action records	4.1.<u>Test plans and procedures</u> are prepared according to requirements. 4.2.Test results are recorded accurately using standard formats. 4.3.Non-conformities and defects are documented with supporting evidence. 4.4.<u>Corrective and preventive actions</u> are recorded and tracked. 4.5.Final test reports are compiled and submitted according to procedures.

Range of Variables

Variable	Range May include but not limited to:
1. Debugging instruments	1.1. Oscilloscope 1.2. Logic Analyzer 1.3. Digital Multimeter 1.4. JTAG/SWD Debugger 1.5. Current Probe/Power Analyzer 1.6. Serial Terminal
2. Logic analyzer captures	2.1. Channel Setup 2.2. Sampling Rate 2.3. Trigger Conditions 2.4. Protocol Analyzers 2.5. Timing Display 2.6. Data Export
3. JTAG/SWD debugging sessions	3.1. Breakpoints 3.2. Watch Variables 3.3. Register View 3.4. Memory View 3.5. Step Execution

Variable	Range May include but not limited to:
	3.6. Flash Programming
4. Protocol decoding functions	4.1. UART Decode 4.2. I2C Decode 4.3. SPI Decode 4.4. PWM Measurement 4.5. 1-Wire Decode 4.6. CAN Decode (Introductory)
5. Fault symptoms	5.1. No Power 5.2. No Communication 5.3. Incorrect Output 5.4. Intermittent Operation 5.5. Reset/Brown-out 5.6. Overheating
6. Troubleshooting techniques	6.1. Visual Inspection 6.2. Signal Tracing 6.3. Divide and Conquer 6.4. Substitution 6.5. Firmware Instrumentation 6.6. Boundary Testing
7. Low-power modes	7.1. Sleep 7.2. Stop 7.3. Standby 7.4. Deep Sleep 7.5. Idle 7.6. Power-down
8. Wake-up sources	8.1. External Interrupt 8.2. RTC Alarm 8.3. Timer 8.4. UART Activity 8.5. Watchdog 8.6. GPIO Event
9. Power profiling measurements	9.1. Current Measurement 9.2. Power Consumption per Mode

Variable	Range May include but not limited to:
	9.3. Duty Cycle Analysis 9.4. Battery Life Estimation 9.5. Energy per Operation 9.6. Peak Current Analysis
10. Test plans and procedures	10.1. Functional Test Plan 10.2. Interface Test Procedure 10.3. Power Test Procedure 10.4. Environmental Checks 10.5. Acceptance Criteria 10.6. Traceability Matrix
11. Corrective and preventive actions	11.1. Root Cause Analysis 11.2. Rework Instructions 11.3. Firmware Fix Records 11.4. Design Change Notes 11.5. Re-test Records 11.6. Preventive Recommendations

Curricular Content Guide:

1. Underpinning Knowledge	1. Operation of oscilloscopes, logic analyzers and JTAG/SWD debuggers. 2. Protocol decoding and communication signal analysis. 3. Systematic fault diagnosis and troubleshooting methods. 4. MCU low-power modes, wake-up mechanisms and clock management. 5. Power profiling techniques and battery life estimation. 6. Test planning, test reporting and documentation standards. 7. Corrective and preventive action recording and tracking.
2. Underpinning Skills	1. Setting up and using debugging instruments 2. Capturing and decoding UART, I2C and SPI signals 3. Debugging firmware using JTAG/SWD tools 4. Diagnosing and correcting hardware-firmware faults 5. Configuring low-power modes and wake-up sources 6. Performing power profiling and optimization

	7. Preparing test reports and corrective action records
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Electronics laboratory 2. Oscilloscopes, logic analyzers and JTAG/SWD debuggers 3. MCU development boards and prototypes 4. Power measurement tools 5. Learning manuals and test templates

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. used oscilloscopes, logic analyzers and JTAG/SWD for debugging 2. decoded protocols and troubleshoot hardware-firmware faults 3. applied low-power design practices and power profiling 4. prepared test reports and corrective action records
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: DEVELOP INDUSTRIAL SMART SYSTEM APPLICATION AND CAPSTONE PROJECT	Nominal Duration: 36 hrs.	Unit Code: SICIP-SEMI-SSEE-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to develop an industrial smart system application and complete a capstone project demonstrating end-to-end embedded system development. It specifically includes the tasks of planning an end-to-end smart system project, developing schematic, PCB, firmware and interface integration, bringing up, testing and demonstrating a prototype system and presenting project evidence and demonstrating industry readiness.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Plan an end-to-end smart system project	1.1. <u>Project requirements</u> are gathered and analyzed according to the application scenario. 1.2.System architecture and block diagrams are prepared. 1.3. <u>Project resources</u> are identified and planned. 1.4.Work breakdown, timeline and milestones are established. 1.5.Project plan is reviewed and approved by the concerned authority.
2. Develop schematic, PCB, firmware and interface integration	2.1.Schematic and PCB design are completed according to project requirements. 2.2. <u>Firmware modules</u> are developed and integrated. 2.3. <u>Sensors, actuators and communication interfaces</u> are integrated into the system. 2.4.Hardware and firmware components are integrated incrementally with testing at each stage. 2.5.Design files and source code are maintained under version control.
3. Bring up, test and demonstrate a prototype system	3.1. <u>Bring-up procedures</u> are performed on the assembled prototype. 3.2.Functional tests are performed against project requirements. 3.3.Faults are diagnosed and corrected during testing. 3.4. <u>System performance parameters</u> are measured and recorded.

Elements of Competency	Performance Criteria
	3.5. Working prototype is demonstrated to assessors and stakeholders.
4. Present project evidence and demonstrate industry readiness	4.1. <u>Project documentation and portfolio</u> are compiled according to standard formats. 4.2. Project outcomes are presented clearly using appropriate media. 4.3. Technical questions are answered confidently and accurately. 4.4. Lessons learned and future improvements are identified. 4.5. <u>Industry readiness competencies</u> are demonstrated during presentation and defense.

Range of Variables

Variable	Range May include but not limited to:
1. Project requirements	1.1. Functional Requirements 1.2. Performance Requirements 1.3. Interface Requirements 1.4. Power Requirements 1.5. Cost Constraints 1.6. Timeline
2. Project resources	2.1. Development Boards 2.2. Components and Modules 2.3. EDA and IDE Software 2.4. Laboratory Instruments 2.5. Budget 2.6. Team Roles
3. Firmware modules	3.1. Driver Modules 3.2. Application Logic 3.3. Communication Handlers 3.4. Data Processing Routines 3.5. User Interface Handlers 3.6. Power Management Routines

Variable	Range May include but not limited to:
4. Sensors, actuators and communication interfaces	4.1. Temperature/Humidity Sensors 4.2. Motion/Level Sensors 4.3. Relays and Motor Drivers 4.4. Display Units 4.5. UART/I2C/SPI Devices 4.6. Wireless Modules (Wi-Fi/Bluetooth)
5. Bring-up procedures	5.1. Visual Inspection 5.2. Power Rail Verification 5.3. Clock and Reset Verification 5.4. Firmware Flashing 5.5. Peripheral Checks 5.6. Communication Checks
6. System performance parameters	6.1. Accuracy 6.2. Response Time 6.3. Power Consumption 6.4. Communication Reliability 6.5. Stability/Uptime 6.6. Range/Coverage (Wireless)
7. Project documentation and portfolio	7.1. Requirement Document 7.2. Schematic and PCB Files 7.3. Firmware Source Code 7.4. Test Reports 7.5. User Guide 7.6. Presentation Slides
8. Industry readiness competencies	8.1. Technical Presentation 8.2. Problem-solving 8.3. Teamwork 8.4. Time Management 8.5. Documentation Quality 8.6. Professional Ethics

Curricular Content Guide:

1. Underpinning Knowledge	1. Project planning, requirement analysis and system architecture design. 2. Integration of schematic, PCB, firmware and application interfaces. 3. Prototype bring-up procedures and systematic testing methods. 4. Performance measurement and evaluation techniques. 5. Documentation, portfolio and presentation standards. 6. Version control and design data management practices. 7. Industry expectations and professional readiness requirements.
2. Underpinning Skills	1. Planning end-to-end embedded smart system projects 2. Developing and integrating schematic, PCB and firmware 3. Integrating sensors, actuators and communication interfaces 4. Bringing up and testing prototype systems 5. Measuring and recording system performance 6. Compiling documentation and presenting project outcomes
3. Underpinning Attitudes	1. Commitment to occupational health and safety 2. Promptness in carrying out activities 3. Sincerity and honesty to duties 4. Environmental concerns 5. Eagerness to learn 6. Tidiness and timeliness 7. Respect for rights of peers and seniors in the workplace 8. Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 1. Project laboratory with complete toolset 2. Development boards, sensors, actuators and modules 3. PCB prototyping facilities 4. Measurement and debugging instruments 5. Documentation templates and presentation facilities

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. planned an end-to-end smart system project
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	2. developed schematic, PCB, firmware and interface integration 3. brought up, tested and demonstrated a prototype system 4. presented project evidence and demonstrated industry readiness
2. Methods of Assessment	Methods of assessment should include: 1. Written test 2. Practical demonstration 3. Oral questioning 4. Portfolio (optional)
3. Context of Assessment	1. Competency assessment must be done in an assessment/training center or in an actual or simulated workplace after completion of the training. 2. Assessment should be done by a certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Smart Systems Engineering: Embedded Design and Semiconductor Application Interfaces
Number of Trainees:	25

Course-wise Training Space:

- Computer Lab: 400 sft
- Electronics Workshop/Laboratory: 700-800 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and Facilities	Required Facilities
1.	Computer/Laptop (Trainer)	01
2.	Printer	01
3.	Multimedia Projector	01
4.	High-performance Desktop/Laptop	25
5.	Embedded IDE and Toolchain Software (Arduino IDE / STM32CubeIDE / PlatformIO or equivalent)	25
6.	EDA/PCB Design Software (Schematic Capture and PCB Layout)	25
7.	MCU Development Boards (STM32 / ESP32 / Arduino / RP2040 or equivalent)	25
8.	Sensor and Interface Module Kits (Temperature, Humidity, Pressure, IMU, Light, Gas, Display, Relay, etc.)	25
9.	Breadboard and Electronic Component Kits	25
10.	JTAG/SWD Debugger/Programmer	25
11.	Digital Multimeter	25
12.	ESD-safe Workbench with ESD Mat and Wrist Strap	25
13.	Digital Storage Oscilloscope	13
14.	Logic Analyzer	13
15.	DC Regulated Power Supply	13
16.	Soldering Station with Fume Extractor	13

S.N.	Major Training Equipment and Facilities	Required Facilities
17.	Function Generator	05
18.	Current Probe/Power Profiling Tool	05
19.	PCB Prototype Fabrication and Assembly Support (In-house or Outsourced)	01 set
20.	Internet Connectivity with Network Switch and Router	01
21.	Central File/Storage Server	01
22.	First Aid and Fire Safety Equipment	01 set

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of the training course on Smart Systems Engineering: Embedded Design and Semiconductor Application Interfaces, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.